

A Geometric Scaling for Fermionic Dark Matter from the Anti-Invariant Lattice of Nikulin Involutions in G_2 Compactifications

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Abstract

We report a sharp numerical correspondence between the topology of K3 surfaces endowed with a Nikulin involution and the mass scale of fermionic dark matter. In twisted connected sum (TCS) compactifications of M-theory on G_2 -holonomy manifolds, the Nikulin involution σ induces a canonical decomposition of the K3 cohomology lattice into invariant (L_+) and anti-invariant (L_-) sublattices. While L_+ (rank 13) has been previously linked to the leptonic CP-violating phase ($\delta_{CP} \approx 197.8^\circ$), we propose a **Geometric Normalization Conjecture (GNC)** where the dark sector mass scale m_χ is determined by the dimensionality of L_- . Our prediction $m_\chi \approx 56.78$ keV matches the optimal fit for fermionic dark matter cores in the Galactic Center (RAR model). This note provides the heuristic dynamical basis for this scaling and its immediate falsifiability via S2 star precession residuals.

Classification: Speculative Research Note (Line B)

Status: Falsifiable by Sgr A* Astrometry (2026–2028)

1 Introduction: The Duality of the Lattice

In G_2 -holonomy compactifications constructed via the twisted connected sum (TCS) method [4,5], the topology is dictated by the Nikulin involution σ acting on the K3 fibers. This involution splits the second cohomology lattice into two orthogonal eigenspaces:

$$H^2(\text{K3}; \mathbb{Z}) \cong L_+ \oplus L_- \tag{1}$$

where L_+ and L_- denote the invariant and anti-invariant sublattices, respectively. Their ranks are topologically fixed as $(L_+) = 13$ and $(L_-) = 9$.

While the invariant sector L_+ governs the visible degrees of freedom (Standard Model charges and CP phases), the anti-invariant “shadow” sector L_- naturally hosts sterile modes, including fermionic dark matter candidates. This dual projection and its associated observables are summarized schematically in Fig. 1.

2 The Geometric Normalization Conjecture (GNC)

We propose that fermionic masses in the shadow sector are not free parameters but arise through a geometric normalization relative to the visible sector’s IR reference scale.

K3 Cohomology with Nikulin Involution	
$H^2(K3; \mathbb{Z})$, rank = 22	
σ -eigenvector decomposition	
Invariant Sector (Light)	Anti-Invariant Sector (Shadow)
L_+ (rank = 13)	L_- (rank = 9)
$\sigma = +1$	$\sigma = -1$
Visible Physics (Line A)	Dark Sector (Line B)
$\delta_{CP} \simeq 197.8^\circ$	$m_\chi \simeq 56.78$ keV
(Test: DUNE)	(Test: S2 / GRAVITY+ / EHT)

Figure 1: **Dual projection of the Nikulin lattice in G_2 compactifications.** The Nikulin involution induces a canonical decomposition of the K3 cohomology lattice. The invariant sector (rank 13) governs visible physics (Line A), while the anti-invariant sector (rank 9) defines a shadow sector whose geometric normalization fixes the fermionic dark matter mass (Line B).

2.1 The Scaling Law

Using the electron mass ($m_e = 511$ keV) as the fundamental normalization scale for the invariant sector—being the lightest charged fermion with a non-zero, well-protected mass—we define the darkino mass m_χ as:

$$m_\chi = \frac{m_e}{(L_-)} = \frac{511 \text{ keV}}{9} \approx 56.78 \text{ keV} \quad (2)$$

This relation suggests that the darkino mass is a topological fragment of the electronic energy scale, distributed across the 9 anti-invariant cycles of the G_2 manifold.

2.2 Heuristic Dynamical Basis

While a first-principles derivation is pending, the GNC is motivated by the following mechanisms:

- **Flux Fractionation:** M-theory G_4 flux distributed over the N independent cycles of L_- suggests a mass scaling $m_\chi \sim \langle G_4 \rangle / (L_-)$.
- **Kinetic Normalization:** In the effective action, the normalization of fermionic kinetic terms over a degenerate lattice of N anti-invariant modes induces a $1/N$ suppression in the effective mass eigenvalues.
- **Instanton Actions:** Non-perturbative contributions to the hidden superpotential from M2-instantons wrapping L_- cycles scale inversely with the volume-weighted rank of the anti-invariant sublattice.

We emphasize that these mechanisms motivate the scaling but do not yet constitute a formal derivation.

3 Astrophysical Convergence at Sgr A*

The **Ruffini-Argüelles-Rueda (RAR)** model [1] identifies a critical mass of $m_\chi \approx 56$ keV for a fermionic core at the Galactic Center. Recent fits to S2 star precession and Milky Way rotation curves [2] confirm this as the “sweet spot” for core-halo stability.

Our geometric prediction, $m_{\chi}^{\text{geo}} = 56.78$ keV, lies within 1.4% of this value. This mass sits slightly above the transition point governing the sign of stellar orbital precession, predicting a **weakly prograde precession** for the S2 star, distinguishable from purely Schwarzschild dynamics. Data from GRAVITY+ and the Event Horizon Telescope (2026–2028) will serve as the definitive empirical test.

4 Conclusion

The Nikulin lattice (rank 22) provides a unified topological map: 13 degrees of freedom for the Light (L_+) and 9 for the Shadow (L_-). The upcoming data from the Galactic Center will determine whether this geometric scaling reflects physical reality.

¹

References

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- [5] A. Corti, M. Haskins, J. Nordström and T. Pacini, “ G_2 -manifolds and associative submanifolds via semi-Fano 3-folds,” *Duke Math. J.* **164**, 1971 (2015).
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¹“Geometry has only the duty to remain consistent across all observables.”